

SafeAlert Database Schema

Complete table documentation with columns, descriptions, constraints, and relationships.

users

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
username	Login name	NOT NULL, UNIQUE, indexed	
email	Email address	UNIQUE, nullable	
password_hash	Hashed password	NOT NULL	
first_name	First name	nullable	
last_name	Last name	nullable	
phone_number	Phone number	nullable	
profile_picture	Profile image path	nullable	
is_department	Department/task allocator flag	NOT NULL, default FALSE	
is_responder	Responder/field worker flag	NOT NULL, default FALSE	
department_id	Department reference	nullable	FK → departments.id
badge_number	Responder badge	nullable	
is_on_duty	On-duty flag	NOT NULL, default FALSE	
is_available	Available for assignments	NOT NULL, default TRUE	
emergency_contacts	Emergency contacts (JSON)	nullable	
medical_info	Medical info (optional)	nullable	
date_joined	Registration time	NOT NULL	
last_login	Last login time	nullable	
updated_at	Last update time	NOT NULL	

categories

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
name	Category name	NOT NULL, UNIQUE, indexed	
description	Category description	nullable	
icon	Icon name or emoji	nullable	
color	UI color code	nullable	
default_severity	Default severity	NOT NULL, default 'MEDIUM'	

is_active	Active flag	NOT NULL, default TRUE	
priority_order	Sort order	default 0	
created_at	Creation time	NOT NULL	
updated_at	Last update time	NOT NULL	

category_department_mappings

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
category_id	Category reference	NOT NULL	FK → categories.id
department_type	Department type (FIRE, POLICE, etc.)	NOT NULL	
priority	Priority (1 = primary)	NOT NULL, default 1	
is_required	Required or optional	default TRUE	

departments

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
name	Department name	NOT NULL, indexed	
code	Department code	NOT NULL, UNIQUE	
type	Department type (FIRE, POLICE, etc.)	NOT NULL, indexed	
description	Description	nullable	
parent_department_id	Parent department	nullable	FK → departments.id
headquarters_lat	HQ latitude	NOT NULL	
headquarters_lng	HQ longitude	NOT NULL	
address	Address	nullable	
coverage_radius_km	Coverage radius (km)	default 15.0	
coverage_polygon	GeoJSON coverage polygon	nullable	
max_concurrent_incidents	Max concurrent incidents	default 5	
current_active_incidents	Current active incidents	default 0	
dispatch_phone	Dispatch phone	nullable	
dispatch_email	Dispatch email	nullable	
operating_hours	Operating hours (JSON)	nullable	
is_24_7	24/7 flag	default TRUE	
is_active	Active flag	NOT NULL, default TRUE	
created_at	Creation time	NOT NULL	

updated_at	Last update time	NOT NULL	
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incident_reports

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
user_id	Reporter	nullable	FK → users.id
is_anonymous	Anonymous report flag	NOT NULL, default FALSE	
anonymous_tracking_code	Tracking code for anonymous reports	nullable, UNIQUE	
category_id	Incident category	NOT NULL	FK → categories.id
severity	Severity (CRITICAL, HIGH, etc.)	NOT NULL, indexed	
title	Title	nullable	
description	Description	NOT NULL	
latitude	Latitude	nullable	
longitude	Longitude	nullable	
status	Status (REPORTED, VERIFIED, etc.)	NOT NULL, indexed	
is_verified	Verified on scene	NOT NULL, default FALSE	
created_at	Creation time	NOT NULL, indexed	
dispatch_time	Dispatch time	nullable	
acknowledge_time	Acknowledgment time	nullable	
arrival_time	Arrival time	nullable	
resolution_time	Resolution time	nullable	
closed_time	Closed time	nullable	
source	Source (WEB, MOBILE, etc.)	NOT NULL, default 'WEB'	
ip_address	IP address	nullable	
user_agent	User agent	nullable	
updated_at	Last update time	NOT NULL	

incident_attachments

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
incident_id	Incident reference	NOT NULL	FK → incident_reports.id
file_path	File path	NOT NULL	
uploaded_at	Upload time	NOT NULL, indexed	

incident_media

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
incident_id	Incident reference	NOT NULL	FK → incident_reports.id
file_path	File path	NOT NULL	
media_type	IMAGE, VIDEO, AUDIO, DOCUMENT	NOT NULL	
mime_type	MIME type	nullable	
thumbnail_path	Thumbnail path	nullable	
captured_at	Capture time	nullable	
uploaded_at	Upload time	NOT NULL, indexed	

incident_messages

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
incident_id	Incident reference	NOT NULL	FK → incident_reports.id
sender_id	Sender	NOT NULL	FK → users.id
message	Message text	NOT NULL	
created_at	Creation time	NOT NULL, indexed	

incident_assignments

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
incident_id	Incident reference	NOT NULL	FK → incident_reports.id
department_id	Department reference	NOT NULL	FK → departments.id
responder_id	Assigned responder	nullable	FK → users.id
priority_rank	Priority rank (1 = highest)	NOT NULL, default 1	
distance_km	Distance (km)	nullable	
allocation_score	Allocation score (0-100)	nullable	
score_breakdown	Score breakdown (JSON)	nullable	
status	ASSIGNED, ACCEPTED, etc.	NOT NULL, indexed	
assigned_at	Assignment time	NOT NULL, indexed	
acknowledged_at	Acknowledgment time	nullable	
en_route_at	En route time	nullable	

arrived_at	Arrival time	nullable	
completed_at	Completion time	nullable	
acknowledgment_time_seconds	Time to accept	nullable	
travel_time_seconds	En route → arrival	nullable	
total_response_time_seconds	Assigned → arrived	nullable	
notes	Notes	nullable	
decline_reason	Decline reason	nullable	
created_at	Creation time	NOT NULL	
updated_at	Last update time	NOT NULL	

resources

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
department_id	Department reference	NOT NULL	FK → departments.id
type	VEHICLE, EQUIPMENT, PERSONNEL	NOT NULL	
name	Resource name	NOT NULL	
identifier	License plate, serial, badge	nullable	
description	Description	nullable	
status	AVAILABLE, DEPLOYED, etc.	NOT NULL, indexed	
current_incident_id	Current incident	nullable	FK → incident_reports.id
current_lat	Current latitude	nullable	
current_lng	Current longitude	nullable	
last_location_update	Last location update	nullable	
capacity	Capacity (e.g. seats)	nullable	
specifications	Additional specs (JSON)	nullable	
created_at	Creation time	NOT NULL	
updated_at	Last update time	NOT NULL	

safety_alerts

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
title	Alert title	NOT NULL	
message	Alert message	NOT NULL	
alert_type	INCIDENT_AREA, WEATHER, etc.	NOT NULL, indexed	

severity	INFO, WARNING, CRITICAL	NOT NULL, indexed	
instructions	Public instructions	nullable	
center_lat	Center latitude	nullable	
center_lng	Center longitude	nullable	
radius_km	Radius (km)	nullable	
coverage_polygon	GeoJSON polygon	nullable	
is_citywide	Citywide flag	NOT NULL, default FALSE	
incident_id	Related incident	nullable	FK → incident_reports.id
active_from	Start time	NOT NULL	
active_until	End time	nullable	
is_active	Active flag	NOT NULL, indexed	
is_expired	Expired flag	NOT NULL, default FALSE	
created_by_id	Creator	nullable	FK → users.id
created_at	Creation time	NOT NULL, indexed	
updated_at	Last update time	NOT NULL	

status_history

Column	Description	Constraints	PK / FK
id	Primary key	NOT NULL	PK
incident_id	Incident reference	NOT NULL	FK → incident_reports.id
old_status	Previous status	nullable	
new_status	New status	NOT NULL	
changed_by_id	User who changed	nullable	FK → users.id
changed_at	Change time	NOT NULL, indexed	
notes	Notes	nullable	
source	API, SYSTEM, ESCALATION	nullable	
assignment_id	Related assignment	nullable	FK → incident_assignments.id